WORKED SOLUTION

STEP III 2012, Question 13

Probability and Statistics • Truncated Normal Distribution

STEP III 2012, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/12-s3-q13

The key idea

A conditional expectation over an interval is an integral of $z\,f(z)$ over that interval, divided by the probability of the interval. For the standard Normal, $z\,\mathrm{e}^{-z^2/2}$ is exactly the derivative of $-\mathrm{e}^{-z^2/2}$, so the integral can be done exactly. Everything after that is standardising, $X = \mu + \sigma Z$, and splitting $|X|$ according to the sign of X.

Solution

Throughout, write $\phi(z) = \dfrac{1}{\sqrt{2\pi}}\,\mathrm{e}^{-z^2/2}$ for the probability density function of Z, so that $\Phi'(z) = \phi(z)$ and $\phi'(z) = -z\,\phi(z)$.

Part (i)

The conditional density of Z given $a < Z < b$ is $\phi(z)/\mathrm{P}(a < Z < b)$ on (a, b), and $\mathrm{P}(a < Z < b) = \Phi(b) - \Phi(a)$. Hence

$$\begin{aligned}
\mathrm{E}(Z \mid a < Z < b) &= \frac{1}{\Phi(b) - \Phi(a)} \int_a^b z\,\phi(z)\,\mathrm{d}z \\
&= \frac{1}{\sqrt{2\pi}\,(\Phi(b) - \Phi(a))} \Big[-\mathrm{e}^{-z^2/2} \Big]_a^b \\
&= \frac{\mathrm{e}^{-a^2/2} - \mathrm{e}^{-b^2/2}}{\sqrt{2\pi}\,(\Phi(b) - \Phi(a))},
\end{aligned}$$

since $\dfrac{\mathrm{d}}{\mathrm{d}z}\left(-\mathrm{e}^{-z^2/2}\right) = z\,\mathrm{e}^{-z^2/2}$. This is the required result. It also holds with $b = \infty$ (or $a = -\infty$), reading $\mathrm{e}^{-\infty} = 0$ and $\Phi(\infty) = 1$, because the integral converges.

Part (ii)

Write $X = \mu + \sigma Z$ with $Z \sim \mathrm{N}(0,1)$ and $\sigma > 0$. Since $\sigma > 0$,

$$X > 0 \iff Z > -\mu/\sigma,$$

so conditioning on $X > 0$ is the same as conditioning on $Z > -\mu/\sigma$. Expectation is linear under conditioning too, so

$$\mathrm{E}(X \mid X > 0) = \mathrm{E}(\mu + \sigma Z \mid Z > -\mu/\sigma) = \mu + \sigma\,\mathrm{E}(Z \mid Z > -\mu/\sigma),$$

as required.

The mean of $|X|$. Put $c = \mu/\sigma$. Part (i) with $a = -c$ and $b = \infty$ gives

$$\mathrm{E}(Z \mid Z > -c) = \frac{\mathrm{e}^{-c^2/2}}{\sqrt{2\pi}\,(1 - \Phi(-c))} = \frac{\phi(c)}{p}, \qquad \text{where } p = \mathrm{P}(X > 0) = 1 - \Phi(-c).$$

Multiplying the previous display by p,

$$\mathrm{E}(X;\, X > 0) := \mathrm{E}(X \mid X > 0)\,\mathrm{P}(X > 0) = \mu p + \sigma\phi(c).$$

Ask yourself How do we get the contribution from $X < 0$ without a second integral?

The two pieces add up to the whole mean: $\mathrm{E}(X;\, X > 0) + \mathrm{E}(X;\, X < 0) = \mathrm{E}(X) = \mu$ (the event $X = 0$ has probability 0). So $\mathrm{E}(X;\, X < 0) = \mu - \mu p - \sigma\phi(c)$. On $X < 0$ we have $|X| = -X$, and so

$$\begin{aligned} m = \mathrm{E}|X| &= \mathrm{E}(X;\, X > 0) - \mathrm{E}(X;\, X < 0) \\ &= 2(\mu p + \sigma\phi(c)) - \mu \\ &= \mu(2p - 1) + 2\sigma\phi(c). \end{aligned}$$

Finally $2p - 1 = 1 - 2\Phi(-\mu/\sigma)$ and $2\phi(c) = \sqrt{2/\pi}\,\mathrm{e}^{-\mu^2/(2\sigma^2)}$, giving

$$m = \mu(1 - 2\Phi(-\mu/\sigma)) + \sigma\sqrt{2/\pi}\,\exp(-\tfrac{1}{2}\mu^2/\sigma^2),$$

as required.

The variance of $|X|$. The point is that $|X|^2 = X^2$, so the second moment needs no new work:

$$\mathrm{E}(|X|^2) = \mathrm{E}(X^2) = \mathrm{Var}(X) + (\mathrm{E}X)^2 = \sigma^2 + \mu^2.$$

Therefore

$$\mathrm{Var}|X| = \mu^2 + \sigma^2 - m^2.$$

Checks

- If $\mu = 0$ then $|X|$ is half-Normal and the formula gives $m = \sigma\sqrt{2/\pi}$, the familiar value; the variance is then $\sigma^2(1 - 2/\pi)$, which is positive.
- If $\mu \gg \sigma$ then X is almost always positive, $\Phi(-\mu/\sigma) \approx 0$ and the exponential term is tiny, so $m \approx \mu$ and $\mathrm{Var}|X| \approx \sigma^2$, as it should be.
- The formula for m is unchanged on replacing μ by $-\mu$ (use $1 - 2\Phi(-c) = -(1 - 2\Phi(c))$), matching the symmetry $|X| \sim |-X|$.
- Numerically, with $\mu = 1$, $\sigma = 2$ we get $m \approx 1.7912$ and $\mathrm{Var}|X| \approx 1.7917$, which agree with direct quadrature and with a simulation of 2×10^5 values.

Common slips

- Forgetting to divide by $\Phi(b) - \Phi(a)$ in part (i): the integral alone is $\mathrm{E}(Z;\, a < Z < b)$, not the conditional expectation.
- Writing $\mathrm{E}|X| = \mathrm{E}(X \mid X > 0)$: the conditional expectations must be weighted by $\mathrm{P}(X > 0)$ and $\mathrm{P}(X < 0)$.
- Computing $\mathrm{E}(X^2)$ by another integral, when $|X|^2 = X^2$ gives it at once.

Beyond STEP

The distribution of $|X|$ is the *folded Normal*, and the conditional distribution in part (i) is a *truncated Normal*; the ratio $\phi(c)/\big(1 - \Phi(-c)\big)$ is the inverse Mills ratio, central to models in econometrics where data are censored or selected. The trick $\phi'(z) = -z\phi(z)$ is also the start of Stein's method for Normal approximation. For more on conditional expectation, see Blitzstein and Hwang, *Introduction to Probability*.

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